BCS-069	INTERNET TECHNOLOGY	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To learn HTML tags and JavaScript Language programming concepts and techniques.
CO2	To develop the ability to logically plan and develop web pages.
CO3	To learn to write, test, and debug web pages using HTML and JavaScript.
CO4	To learn to write, test, and debug web pages using servlet
CO5	To learn to write, test, and debug web pages using JSP.

MODULE-I

Introduction and Web Development Strategies: History of Web, World Wide Web Consortium (W3C), Protocols governing Web, Creating Websites for individual and Corporate World, CyberLaws, Web Applications, Writing Web Projects, Identification of Objects, Target Users, Web Team.
HTML & DHTML: Introduction, Standard Structure, Basic html tags such as list, table, images, frames, forms. Cascading Style Sheet (CSS).**XML:** Introduction & use, DTD, XML Schema.**Object Model:** Document Object Model (DOM), Simple API for XML (SAX).Introduction to Java Script, Object in JavaScript, Introduction to AJAX.

MODULE-II

Server Side Programming: Introduction to web Server, Configuring the Tomcat Web Container.
The Servlet Technology: Introduction to Servelets, Lifecycle, The javax.servlet Package, Obtaining Configuration Information, Preserving the ServletConfig , The Servlet Context , Sharing Information Among Servlets, Requests and Responses , The GenericServlet Wrapper Class , The HttpServlet Class, The HttpServletRequest Interface , HttpServletResponse , Sending an Error Code , Sending Special Characters , Buffering the Response , Populating HTML Elements , Request Dispatching , Creating Thread-Safe Servlets.
The JSP Technology: What's Wrong with Servlets? , Running Your First JSP , How JSP Works , The JSP Servlet Generated Code , The JSP API , The Generated Servlet Revisited , Implicit Objects. Directives, Scripting Elements, Standard Action Elements, Comments, Converting into XML Syntax.
Session Management: What Is Session Management? , URL Rewriting , Hidden Fields , Cookies , Session Objects , Knowing Which Technique to Use.

MODULE-III

Database Connectivity: Database Programming using JDBC, Studying Javax.sql.*package, accessing a database from a JSP page, Application-specific Database Action, Developing Java Beans in a JSP page, introduction to Struts framework.

Text Books:

1. "Java for the Web with Servlets, JSP, and EJB: A Developer's Guide to J2EE Solutions", Budi Kurniawan , New Riders Publishing.
2. Burdman, "Collaborative Web Development" Addison Wesley.

Reference Books:

1. Chris Bates, "Web Programing Building Internet Applications", 2nd Edition, WILEY, Dreamtech
2. Joel Sklar , "Principal of web Design" Vikash and Thomas Learning
3. Horstmann, "CoreJava", Addison Wesley.
4. Herbert Schieldt, "The Complete Reference:Java", TMH.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Explain the concepts of internetworking techniques with their characteristics.
CO2	Illustrate the requirement for WWW format and techniques.
CO3	Define the email functioning and basics of HTML,XML and DHTML.
CO4	Recognize the functioning of servers and privacy, security related mechanism.
CO5	Differentiate between different Web Extensions and Web Services.